APPROVED ASSEMBLIES

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 1I: Wood, Insulated

Deck Description: $^{19}/_{32}$" or greater plywood or wood plank

System Type C(1): All layers of insulation simultaneously attached; membrane fully adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft²
InsulBase, StormBase, H-Shield NB, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5" thick	**1, 3, 4**	**1:2 ft²**
Minimum 2" thick	**1, 5, 6**	**1:4 ft²**
Structodeck High Density Wood Fiber Board		
Minimum ½" thick	**3, 4**	**1:2 ft²**

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: ¼", $^3/_8$", ½", or $^5/_8$" gypsum.

Membrane #1: Sure-Weld, Sure-Weld HS, Reinforced, 45 or 60 mil membrane or Sure-Weld EXTRA, 72 or 80 mil membrane fully adhered to the insulation using Sure-Weld Bonding Adhesive, or LVOC Bonding Adhesive applied to the substrate at a rate of 1 gal/60 ft.² (finished surface) or Aqua Base 120 Bonding Adhesive applied to the substrate at a rate of 1 gal/120 ft.²
Maximum Design Pressure –45 psf (See General Limitation #9)

Membrane #2: Sure-Weld FleeceBACK 100 or 115 mil membrane fully adhered to the insulation using Aqua Base 120 Bonding Adhesive applied to the substrate at a rate of 1 gal/120 ft.²
Maximum Design Pressure –45 psf. (See General Limitation #9)

Sure-Weld FleeceBACK 100 or 115 mil membrane fully adhered to the insulation using Flexible FAST Adhesive applied to the substrate at a rate of 1 gal/sq.
Maximum Design Pressure –45 psf. (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 1I: Wood, Insulated

Deck Description: $^{19}/_{32}$" or greater plywood or wood plank

System Type C(2): Membrane adhered over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer (Optional)	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD, H-Shield HD Composite		
Minimum ½-inch thick	**N/A**	**N/A**

Top or Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SECUROCK Gypsum-Fiber Roof Board		
Minimum 1/2-inch thick	**3, 8**	**1:2 ft²**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane fully adhered to the insulation using Sure-Weld Bonding Adhesive or Sure-Weld Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area.

Membrane Splices: Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld along the outside edge.

Maximum Design Pressure: -75 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, Fleecebacked

Deck Type 1I: Wood, Insulated

Deck Description: $^{19}/_{32}$" or greater plywood or wood plank

System Type D(1): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5" thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof

Barrier: (Optional)¼", $^3/_8$", ½", or $^5/_8$" gypsum.

Membrane: Sure-Weld, Sure-Weld HS, Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening #1: 6-foot wide Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA membrane mechanically fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates.
Maximum Design Pressure -67.5 psf. (See General Limitation #7)

Fastening #2: 8-foot wide Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA membrane mechanically fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates.
Maximum Design Pressure -52.5 psf. (See General Limitation #7)

Membrane Splices: Overlap membrane splices a minimum of 5-1/2 inches to provide for a minimum 1-1/2 inch heat weld along the outside edge.

Maximum Design Pressure: See Membrane Options Above

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 1I: Wood, Insulated

Deck Description: $^{19}/_{32}$" or greater plywood or wood plank

System Type D(2): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer (Optional)	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD, H-Shield HD Composite		
Minimum ½-inch thick	**N/A**	**N/A**

Top or Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SECUROCK Gypsum-Fiber Roof Board, Dens Deck Prime		
Minimum 1/4-inch thick	**N/A**	**N/A**
SecurShield		
Minimum ½-inch thick	**N/A**	**N/A**
SecurShield HD Composite)		
Minimum 2-inch thick	**N/A**	**N/A**
StormBase		
Minimum 1-1/2-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft.

Membrane: Sure-Weld or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening #1: 8-foot wide Sure-Weld (60-mil) or Sure-Weld EXTRA membrane mechanically fastened 12-inches on center through the field seams using HP-Xtra Fasteners and Piranha XTRA Plates.

Membrane Splices: Overlap membrane splices a minimum of 5-1/2 inches to provide for a minimum 1-1/2 inch heat weld along the outside edge.

Maximum Design Pressure: -45 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 1I: Wood, Insulated

Deck Description: $^{19}/_{32}$" or greater plywood or wood plank

System Type D(3): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Any approved insulation layer listed in Table 2 Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft.

Membrane: Sure-Weld or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening #1: 5-foot wide Sure-Weld or Sure-Weld EXTRA membrane mechanically fastened 6-inches on center through the field seams into the rafters spaced 48-inches on center using HP-X Fasteners and Piranha Plates.

Membrane Splices: Overlap membrane splices a minimum of 5-1/2 inches to provide for a minimum 1-1/2 inch heat weld along the outside edge.

Maximum Design Pressure: -75 psf (See General Limitation #7)